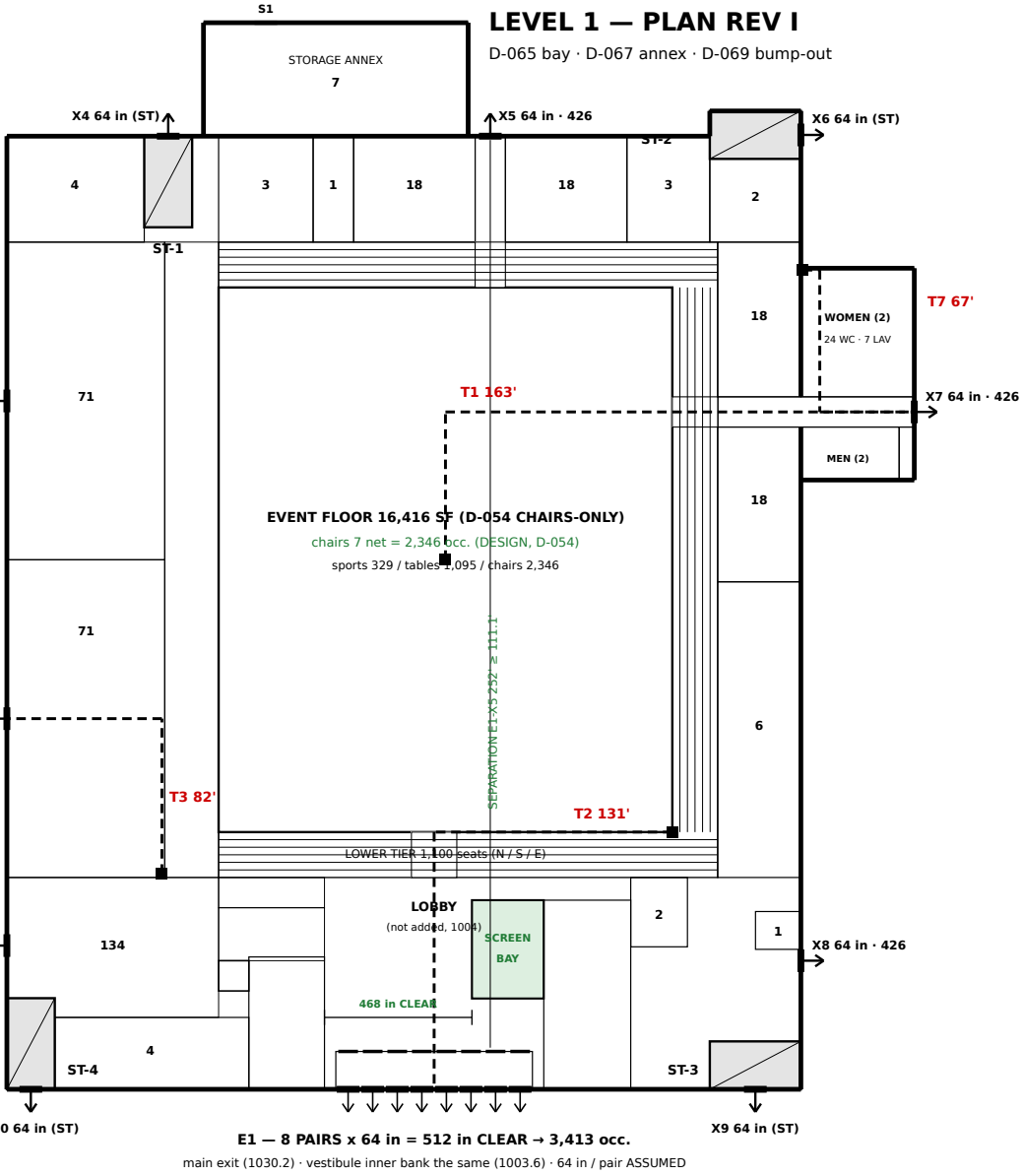




L1 EXITS — CAPACITY (0.15 in / occ., 1005.3.2 exc.)

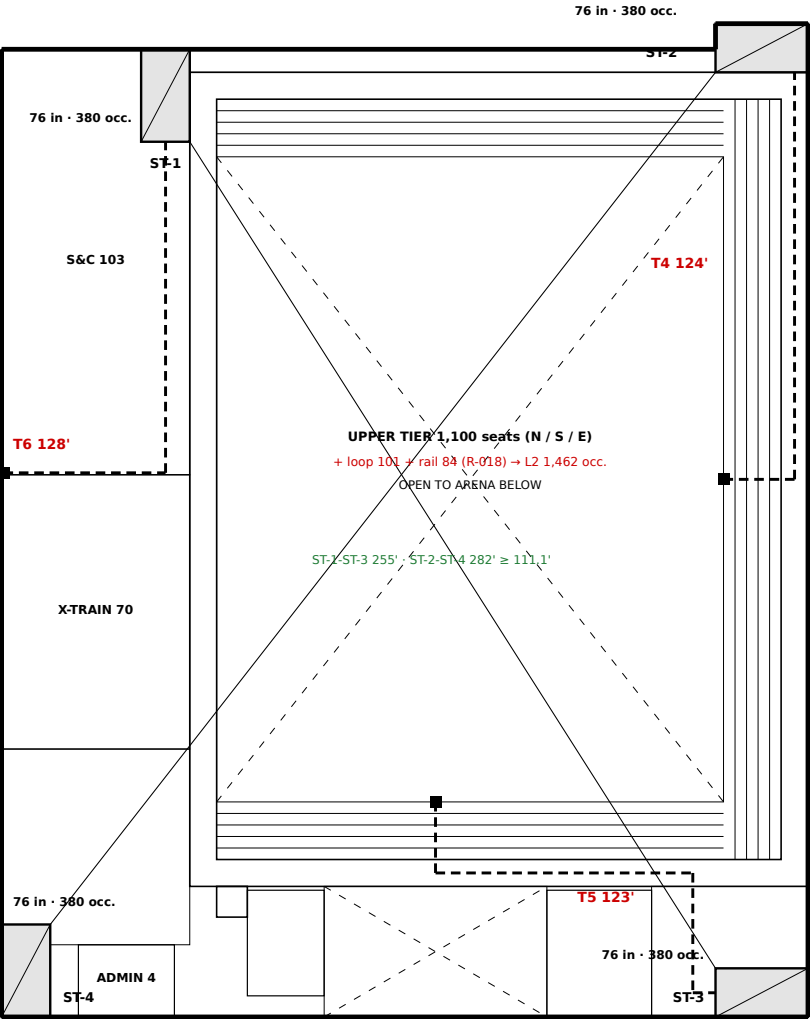
EXIT / CHECK	CLEAR in	CAPACITY	SERVES / RESULT
E1 main (bank)	512	3,413	lobby, portal, south tier, concourses
X1, X2, X3, X5, X7, X8	6 x 64 = 384	426 each	exit-only doors (64 in pair ASSUMED)
X4, X6, X9, X10	4 x 64 = 256	4 x 9.2 in left	stair discharge: L2 uses 54.8 in each
Total, building worst 5,289	need 793.4	have 1,152.0	PASS
Main exit ≥ 1/2 (1030.2)	need 396.7	have 512.0	PASS
Other L1 exits ≥ 1/2 of L1 3,827 (1030.3)	need 287.0	have 420.7	PASS
Lose E1, ≥ 50 % (1005.5)	need 396.7	have 640.0	PASS
Standing margin 6,227: E1 (exits as drawn)	need 467.0	have 512.0	PASS
Lobby beside the screening bay (1003.6)	need 467.0	have 468.0	PASS

TBD

- Door widths, hardware and the 10 ft open space / street frontage at E1 (1030.2): architect. Seating aisles, rails, smoke-protected seating (1030.6.2): TBD. Structure 1'-6" ASSUMED (zero margin at 7'-6").

LEVEL 2 — LIFE SAFETY (PLAN REV F, UNCHANGED)

1" = 50'-0" · 4 stairs 76 in (D-052) = the L2 exits · loop = upper concourse on event days



L2 EXITS — STAIRS (0.2 in / occ., 1005.3.1 exc. 1)

EXIT / CHECK	CLEAR in	CAPACITY	SERVES / RESULT
ST-1 ... ST-4	4 x 76 = 304	380 each	one per corner, discharge X4 / X6 / X9 / X10
Total, L2 worst 1,462	need 292.4	have 304.0	PASS (+11.6)
Lose one stair, ≥ 50 % (1005.5)	need 146.2	have 228.0	PASS (+81.8)
Loop at the worst corner (R-018.3, 0.2)	need 78.6	have 84.0	PASS (+5.4)
4 exits from L2 (T1006.3.3, > 1,000)		4 stairs	PASS

EXIT (E) THROUGH THE BUMP-OUT (D-069) — CHECK

EXIT (E) runs 30 ft on through the bump-out to X7: 8 ft = 96 in ≥ 44 in corridor (T1020.3) and the 64 in X7 pair; restroom doors and the DF alcove open off it, no queuing in it (1003.6). T1 163 ft ≤ 250. X7 discharges east: path to a public way TBD (1028, D-006; C-101 Rev E).

Sources: P2-G-002 Rev B; P2-A-101 Rev I / A-102 Rev F (params/phase2_plan_rev_i.yaml); params/phase2_life_safety_rev_c.yaml; IBC 2021 1003.6, 1006.2.1, 1007.1, 1017.2-3, T1020.3, 1028, 1030.2-3, 2902.3.3; Shane 1:21-1:22 + 3:31 PM CT.

Schematic life safety diagram for the architect of record and the AHJ. Not a code review and no statement of approval; every line needs architect + AHJ confirmation (D-008). Door widths ASSUMED, travel paths measured on the diagram.

LEGEND

Heavy bar + arrow = exit door (clear in · occupants); E1 = 8 pair bars, outer + inner (vestibule) bank. Grey box = stair. Red dashed + dot = travel path (dot = remote point; feet, rectilinear). Green line = exit separation. Green zone = screening bay (D-065). Numbers in rooms = occupant load (P2-G-002).

OCCUPANT LOADS (P2-G-002 REV B, IBC T1004.5)

CASE	FLOOR	L1	BLDG	E1 NEED
Sports (50 gross)	329	1,810	3,272	245.4
Tables + chairs (15 net)	1,095	2,576	4,038	302.8
Chairs only (7 net)	2,346	3,827	5,289	396.7
Standing (5 net) (WORST)	3,284	4,765	6,227	467.0
Whole floor zone 17,280 SF standing	2,469	3,950	5,412	405.9

L1 without the floor 1,481 (seats 1,100); L2 1,277 base, 1,462 worst. All cases pass with the 512 in bank; the whole-floor sensitivity still passes (E1 405.9 ≤ 512 in, total 811.8 ≤ 1,152 in).

EXIT SEPARATION (1007.1.1 exc. 2, sprinklered)

- Max overall diagonal 333.2 ft (SW corner to NE stair tower) → 1/3 = 111.1 ft (1/2 = 166.6 ft unsprinklered). Measured to any point of a doorway / closest riser (1007.1.1.1). All outline corners checked: bump-out NE 323.6 ft, annex NE 307.3 ft (shorter).
- L1 E1 ↔ X5: 252.0 ft ≥ 111.1 ft — PASS. E1 measured at the point of the bank (x 87.25-138.75) in line with X5.
 - L2 ST-1 ↔ ST-3: 255.1 ft ≥ 111.1 ft — PASS. nearest corners of the stair rectangles (risers lie inside, so the riser distance is not shorter).
 - L2 ST-2 ↔ ST-4: 281.5 ft ≥ 111.1 ft — PASS. nearest corners of the stair rectangles.
 - Other exits a reasonable distance apart (1007.1.2): one stair per corner, exits on all four walls.

TRAVEL DISTANCE (Table 1017.2: 250 FT)

IBC 2021 Table 1017.2: Group A with sprinklers (903.3.1.1) 250 ft. 1017.3: from the most remote point along the natural and unobstructed path to the entrance to the nearest exit. Paths below are rectilinear along open floor on the diagram (walls / doors / aisles not drawn): UNVERIFIED.

- T1 L1 event floor center → V2 → EXIT (E) + corridor → X7 (moved 30 ft east): 163 ft — ≤ 250, PASS (UNVERIFIED)
- T2 L1 floor SE corner → portal → lobby → E1: 131 ft — ≤ 250, PASS (UNVERIFIED)
- T3 L1 girls locker far corner → X2: 82 ft — ≤ 250, PASS (UNVERIFIED)
- T4 L2 E upper tier, front row mid → aisle → loop → ST-2: 124 ft — ≤ 250, PASS (UNVERIFIED)
- T5 L2 S upper tier, front row mid → loop → upper concourse (E) → ST-3: 123 ft — ≤ 250, PASS (UNVERIFIED)
- T6 L2 S&C far corner → ST-1: 128 ft — ≤ 250, PASS (UNVERIFIED)
- T7 L1 women's room (2) far corner → corridor → X7: 67 ft — ≤ 250, PASS (UNVERIFIED)

L2 paths end at the stair. If the stairs are open exit access stairways rather than enclosed exit stairways (TBD, architect), travel continues down the stair (1017.3.1, along the nosings) to the L1 exit door: about +50 ft ASSUMED (2 flights + 76 in landings + 5 ft to the door) — still under 250 ft.

COMMON PATH (T1006.2.1, 1030.8)

- Restroom core 2 (D-069) (accessory): one entry each, off the corridor: women far corner → door 38 ft, men 34 ft (≤ 75), then two ways: W to V2 / floor, E to X7 — OK as planned (doors ASSUMED, UNVERIFIED).
- Storage annex (D-067) (7): ≤ 49 and 70 x 30 ft: S1 + the door to equipment storage = 2 ways out; farthest point ≤ 75 ft to either — OK as planned (doors ASSUMED).
- Event lockers 1-4 (18 each): one door OK if ≤ 75 ft: room 32 x 28 / 22 x 41 ft → ≤ 63 ft rectilinear to any wall point — LIKELY — door not located (UNVERIFIED).
- Boys / girls locker (71 each): > 49 → 2 exits: X2 / X3 + corridor door — OK as planned (corridor door ASSUMED).
- Flex / team assembly (134): > 49 → 2 exits: X1 + lobby side (CD-1 free egress, 1010.2) — OK as planned.
- S&C / cross-training (L2) (103 / 70): > 49 → 2 exit access doorways each (ST-1 / loop; corridor / loop) — ASSUMED doors (TBD).
- Seats (both tiers) (1,100 + 1,100): 30 ft to a choice of two paths: aisle 12-15 ft to the floor / loop, then two ways — TBD — aisles and row lengths not drawn.

Rules: one exit only if ≤ 49 occ. and ≤ 75 ft (T1006.2.1, Group A, sprinklered); seats 30 ft to a choice of two paths (1030.8); unoccupied mech rooms exempt (1006.2.1 exc. 3).

SCREENING BAY vs EGRESS (D-065 DECIDED) — CHECK

- Bay x 123-142, y 24-50 (19 x 26 ft); lobby clear beside it 39 ft = 468 in, free of lanes, stanchions and queues; 14 ft cross aisle to the inner bank. Main exit need: chairs 396.7 in (design), standing 467.0 in (margin): both ≤ 468, PASS.
- Lanes, tables, queues and control lines stay out of the 468 in path (IBC 1003.6, 1010.5). Standing is not a design use (D-054, posted loads 1004.9): whole-floor standing would need 479.9 in here.

RESTROOMS — CORE 2 BUMP-OUT (D-069 DECIDED, OPTION 2)

D-069 DECIDED Option 2 (Shane 3:31 PM CT): core 2 in a one-storey 30 x 56 ft bump-out on the east wall at V2 / EXIT (E). Lobby core 10 / 18 WC, 4 / 5 lav, 2 DF + core 2 men 5 WC + 7 urinals (≤ 67 %, IPC 424.2), 5 lav; women 24 WC, 7 lav; 2 DF = 22 / 42, 9 / 12, 4 = chairs-only 3,446 (T2902.1): PASS. Within one storey and 500 ft (2902.3.3): PASS.

- Floor chair layouts must keep an aisle to V2: core 2 is reached only through V2 / EXIT (E) (next A-103 rev).

EXIT DISCHARGE — 40 FT WALK (D-066 DECIDED) — CHECK

Exit discharge (1028.3, D-066 DECIDED): walk 40 ft at the doors = 28 ft brick + 2 x 6 ft bands = 480 in ≥ 466.5 (standing margin 467.0); at the portal the bands pass outside both 8 ft piers (6 ft each) -> 28 + 6 + 6 = 40 ft = 480 in; south to the public way 40 ft (site TBD, D-006). P2-C-101 Rev D.

PRELIMINARY — CONCEPTUAL DESIGN — NOT FOR CONSTRUCTION

PROJECT	PHASE	SHEET TITLE	SCALE	DATE	REV	DRAWN BY	SHEET
Hazel Green Regional Athletic Complex Trojan Horse Arena · Hazel Green area, Madison County, Alabama (site TBD, D-006)	PHASE 2	LIFE SAFETY PLAN LEVELS 1 + 2 · SCHEMATIC	1" = 50'-0"	October 2026	C	Drawn by KEYSTONE (AI) for Shane Brazelton	P2-A-111